GCSE · OCR · Computer Science

2 hours 25 questions

Operating Systems & Utility Software

Operating Systems & Utility Software

The Purpose & Functionality of Operating Systems / The Purpose & Functionality of Utility Software

Easy (3 questions)	/6
Medium (12 questions)	/44
Hard (10 questions)	/47
Total Marks	**/97**

Scan here to return to the course
or visit savemyexams.com

© 2026 Save My Exams, Ltd.

Easy Questions

1 **State two functions** of the operating system.

.....

..... **(2 marks)**

2 **Identify one advantage** of a command line interface.

..... **(1 mark)**

3 **For each statement**, **identify** if the statement is **True or False.**

Statement	True/False?
Utility software is designed to perform a number of tasks	
Utility software interacts with the computers hardware	
Some utility software comes installed with the operating system	

(3 marks)

© 2026 Save My Exams, Ltd.

Medium Questions

1 (a) Layla uses utility system software.

State the purpose of **utility system software**

..... **(1 mark)**

(b) Layla uses **a backup utility**. She performs both full and incremental backups.

Explain the reasons why Layla performs **both full** and **incremental** backups.

.....

.....

..... **(3 marks)**

2 Ali's tablet computer has an operating system.

Complete the following description of the functions of an operating system **by selecting the appropriate missing words from those in the box**.

user, drivers, directories, hardware, interface, multitasking, output, peripherals, printers, processor, RAM, utility, ROM, running, passwords, faster, volatile, virtual.

The operating system provides a user ..... This displays the output to the user and allows the user to interact with the .....

The operating system controls the movement of data from secondary storage to .....and vice-versa. This is known as memory management.

The operating system can only perform one process at a time, but by managing the memory the computer can appear to be completing more than one process at a time. This is known as .....

An operating system allows device .....to be installed to allow an external piece of hardware to interact with the .....

The operating system provides security through user accounts and .....
It also creates and maintains a file system to organise files and .....

(8 marks)

3 Xander's tablet computer comes with system software, including an operating system and utility system software.

The operating system provides file management.

Identify three ways that Xander can **make use** of the **file management facility**.

.....

.....

..... **(3 marks)**

4 (a) The utility system software **provides compression software**. Xander uses this to **compress an image**.

Explain how the compression software **will compress the image** file.

.....

.....

.....

..... **(4 marks)**

(b) **Give the name** of **two** other **types** of utility system software

.....

..... **(2 marks)**

5 (a) William's computer has utility programs installed including automatic backup.

William can **choose between** a **full** or **incremental backup**.

Identify the backup method William **should use** to backup the computer, **justifying** your choice.

.....

.....

..... **(3 marks)**

(b) **Give one additional** utility program William could make use of and **describe** how he would use it.

.....

.....

..... **(3 marks)**

6 A computer has an operating system and utility software.

The table contains operating system functions and a task that each function performs.

Complete the table by writing the two missing function names and a task performed by the two given functions.

Function	Task
	Moves data from secondary storage to RAM
Peripheral management	
	Allows the user to create, name and delete folders
User interface	

(4 marks)

7 **Describe** two differences between a Graphical User Interface (GUI) and a Command Line Interface (CLI)

.....

..... **(2 marks)**

8 **Explain** one advantage of using a Natural Language Interface (NLI) for a smart home device

.....

..... **(2 marks)**

9 **Describe** how the operating system allows **multitasking** to occur

.....

..... **(2 marks)**

10 **Describe** two functions of User Management

.....

..... **(2 marks)**

11 **Identify** three tasks provided by file management software

.....

.....

..... **(3 marks)**

12 **"A student connects a specialised MIDI keyboard to their computer to record music."**

Describe how the **device driver** enables this interaction

.....

..... **(2 marks)**

Hard Questions

1 (a) A restaurant has **a computer-based ordering system** which is **running slowly**. A technician has said that the hard disc drive is **fragmented**. The technician has suggested using utility software to defragment the drive.

Explain how the restaurant's **hard disc could have become fragmented**.

.....

.....

.....

..... **(4 marks)**

(b) **Explain how** defragmentation software could overcome the issue of the slow computer system.

.....

.....

..... **(3 marks)**

2 Layla uses her computer to create educational games

Layla makes use of system software.

One type of system software is the operating system.

Identify and **describe two functions** of an operating system.

.....

.....

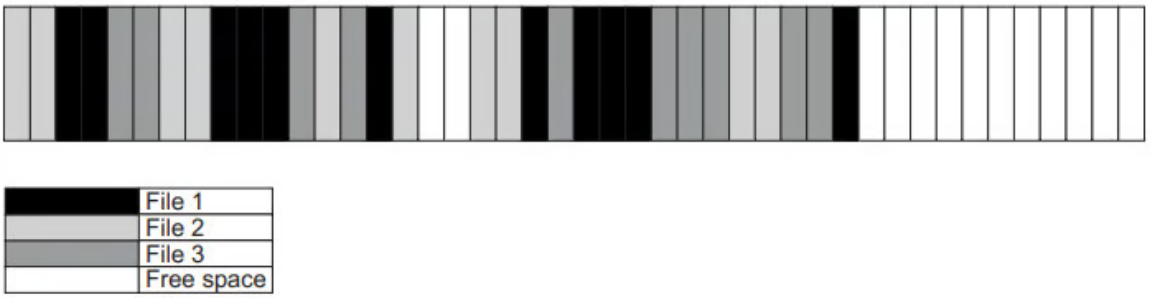
.....

.....

.....

..... **(6 marks)**

3 (a) Ali runs **defragmentation analysis** on his **magnetic hard disk**. Parts of the results are shown.



Explain how defragmentation will **change how the files and free space are arranged** on Ali's hard disk.

.....

.....

..... **(3 marks)**

(b) **After defragmentation**, Ali's computer **is able t**o **access files faster**.

Explain why Ali's computer can access the files faster after defragmentation.

.....

..... **(2 marks)**

(c) **Give three** additional **examples** of utility programs.

.....

.....

..... **(3 marks)**

4 **Complete** the description of utility system software using the words provided in the list. Not all words are used.

- access
- amount
- apart
- compression
- consecutive
- defragmentation
- deleted
- encryption
- key
- lock
- quantity
- separate
- speed
- understood

..... software changes data using a ..... . If the changed data is intercepted, it cannot be ..... . This software does not stop the data from being intercepted. ..... software analyses the data on a disk to find files that have been split and stored in separate locations. The split files are moved to be ..... in storage and the free space is moved together. This does not provide more storage space on the disk, instead it makes the ..... of the data faster because the read head does not have to move as far to access the next part of the file.

(6 marks)

5 **Explain** why a professional systems administrator might prefer a command line interface (CLI) over a graphical user interface (GUI) for automating repetitive tasks

.....

.....

..... **(3 marks)**

6 **Evaluate** the impact of a graphical user interface (GUI) on the performance of a low-powered embedded system

.....

.....

.....

..... **(4 marks)**

7 **Explain** the role of the operating system (OS) when the primary storage (RAM) becomes full

.....

.....

.....

..... **(4 marks)**

8 **Explain** how user management enhances the security of a school network

.....

.....

..... **(3 marks)**

9 **Explain** the importance of a search facility within an operating system

.....

.....

© 2026 Save My Exams, Ltd.

..... **(3 marks)**

10 **Explain** two benefits and one drawback to why a professional designer would choose to install a **manufacturer-specific driver** for a new graphics tablet instead of relying on the **generic driver** provided by the Operating System.

.....

.....

..... **(3 marks)**